Jatinder Sahota

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Education

B.Sc. Computer Science – Mathematics Joint Honours (Co-op) 08/22 – 01/28

University of Manitoba *Winnipeg, MB*

- Minor: Statistics
- GPA: **4.03**/4.50
- Awards: Dean's Honour List, Undergraduate Student Research Award (2x), Best Overall Game – CSSA Game Jam 2025

Technical Skills

- **Programming:** C/C, Java, Python, TypeScript, JavaScript, Julia, BASH, SQL
- **Frameworks & Libraries:** .NET, Angular, JavaFX, Flask, Astro.js, Tailwind CSS
- **Tools:** Git, Postman, Linux, Neovim

Relevant Experience

Undergraduate Student Research Award 05/26 – Present

Department of Mathematics, University of Manitoba *Winnipeg, MB*

Supervised by Dr. Mikael Slevinsky

- Write recursive algorithms in **Julia** to compute fast differentiation matrices between orthogonal polynomial bases.
- Prove algorithm correctness analytically through matrix analysis, numerical analysis, and approximation theory.
- Develop examples demonstrating conversions between orthogonal polynomial families on the Cassini oval.

Full Stack Software Developer 01/26 – 04/26

Conquest Planning *Winnipeg, MB (Remote)*

- Modified **Angular** and **TypeScript** components to implement business logic, resolve UI defects, and integrate with updated **C#/.NET** APIs.
- Extended **C#/.NET** APIs and financial engine logic to support corporation structures, ownership relationships, and plan summary calculations.
- Maintained **95%+** unit test coverage and validated production changes through unit, API (Postman), manual, and end-to-end testing.

NSERC Undergraduate Student Research Award 05/25 – 08/25

Department of Computer Science, University of Manitoba *Winnipeg, MB*

Supervised by Dr. Stephane Durocher, GADA Lab

- Designed and analyzed an algorithm for the Online Traveling Salesman Problem on the circle using theoretical analysis and computational testing.
- Developed and evaluated algorithm optimality through mathematical proofs and extensive literature review.
- Documented methodology and findings in a comprehensive technical report.

Projects

Formula 1 Database System 09/25 – 12/25

- Designed and implemented a **15-table** relational database using **SQLite**, **SQL Server**, and **Java**, normalizing over **100,000** Formula 1 records dating back to **1950** to BCNF/3NF.
- Developed a **Java** business layer responsible for input validation, database communication, pagination, and formatting query results for a command-line REPL.
- Implemented prepared statements to prevent SQL injection while supporting secure CRUD operations and interactive database queries.

Unix-style Shell and exFAT File System Reader 05/25

- Developed a Unix-style shell in **C** supporting command execution, pipes, and input/output redirection.
- Implemented an **exFAT** filesystem reader with volume mounting, file descriptor management, and file tracking using an Open File Table.
- Parsed boot regions, FAT chains, and directory entries to support execution of GNU utilities on mounted filesystem images.